

KEAMANAN JARINGAN



PENYUSUN:

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KELAS : 4SKJA-C

UNIVERSITAS DIPA MAKASSAR

SISTEM INFORMASI

2026

Buatlah Enkripsi dan Deskripsi nama masing-masing, ketentuan:

- a. Menggunakan Kunci Public
- b. Menggunakan Kunci Private

Penyelesaian:

1. Proses pembuatan kunci

- Pilih dua bilangan prima:

$$P = 101 ; q = 113$$

- Nilai n

$$n = p \times q = 101 \times 113 = 11413$$

- Nilai m

$$m = (p - 1)(q - 1) = (101 - 1)(113 - 1) = 100 \times 112 = 11200$$

- Pilih nilai e yang relatif prima terhadap m

$$e = 3$$

- Nilai d

$$ed \equiv 1 \pmod{m}$$

$$3d \equiv 1 \pmod{11200}$$

$$3d = 1 + 11200k \text{ (hasilnya harus bulat)}$$

$$d = \frac{1 + 11200(2)}{3} = \frac{22401}{3} = 7467$$

- Kunci Publik (PK)

$$PK = (e, n) = (3, 11413)$$

- Kunci Private (SK)

$$SK = (d, n) = (7467, 11413)$$

2. Ubah Plaintext ke kode ASCII

BUNGA MAULIDYA JELITHA S

66 85 78 71 65 32 77 65 85 76 73 68 89 65 32 74 69 76 73 84 72 65 32 83

Dalam kasus ini pesan dibagi menjadi blok 4 digit

- $p1 = 6685$
- $p2 = 7871$
- $p3 = 6532$
- $p4 = 7765$
- $p5 = 8576$
- $p6 = 7368$
- $p7 = 8965$
- $p8 = 3274$
- $p9 = 6976$
- $p10 = 7384$
- $p11 = 7265$
- $p12 = 3283$

Dimana setiap blok tidak lebih dari $n - 1 = 11413 - 1 = 11412$

3. Proses Enkripsi

Persamaan	Proses	Nilai C
$P1^e \bmod n$	$6685^3 \bmod 11413$	5041
$P2^e \bmod n$	$7871^3 \bmod 11413$	1475
$P3^e \bmod n$	$8965^3 \bmod 11413$	5265
$P4^e \bmod n$	$7765^3 \bmod 11413$	4635
$P5^e \bmod n$	$8576^3 \bmod 11413$	4826
$P6^e \bmod n$	$7368^3 \bmod 11413$	6743
$P7^e \bmod n$	$6532^3 \bmod 11413$	3251
$P8^e \bmod n$	$3274^3 \bmod 11413$	10256

Persamaan	Proses	Nilai C
$P9^e \bmod n$	$6976^3 \bmod 11413$	3171
$P10^e \bmod n$	$7384^3 \bmod 11413$	220
$P11^e \bmod n$	$7265^3 \bmod 11413$	10060
$P12^e \bmod n$	$3283^3 \bmod 11413$	442
didapat	504114753251463548266743526510256317122010060442	

50 41 14 75 32 51 46 35 48 26 67 43 52 65 10 25 63 17 12 20 10 06 04 42

2)[SO]K[SP]3.#0[SUB]C+4A[LF][EM]?[DC1][FF][DC4][LF][ACK][EOT]*

4. Proses Deskripsi

Persamaan	Proses	Nilai C
$C1^d \bmod n$	$5041^{7467} \bmod 11413$	6685
$C2^d \bmod n$	$1475^{7467} \bmod 11413$	7871
$C3^d \bmod n$	$5265^{7467} \bmod 11413$	8965
$C4^d \bmod n$	$4635^{7467} \bmod 11413$	7765
$C5^d \bmod n$	$4826^{7467} \bmod 11413$	8576
$C6^d \bmod n$	$6743^{7467} \bmod 11413$	7368
$C7^d \bmod n$	$3251^{7467} \bmod 11413$	6532
$C8^d \bmod n$	$10256^{7467} \bmod 11413$	3274
$C9^d \bmod n$	$3171^{7467} \bmod 11413$	6976
$C10^d \bmod n$	$220^{7467} \bmod 11413$	7384
$C11^d \bmod n$	$10060^{7467} \bmod 11413$	7265
$C12^d \bmod n$	$442^{7467} \bmod 11413$	3283
didapat	668578716532776585767368896532746976738472653283	

66 85 78 71 65 32 77 65 85 76 73 68 89 65 32 74 69 76 73 84 72 65 32 83

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(Sumber ASCII : [ASCII table - Table of ASCII codes, characters and symbols](#))